Complete manuscript comparison

Overview v15: two-source Representation

There are 18 topics and 48 recovered originals among 106 accessible originals; 58 remain pending and 55 earlier indexed originals remain archived. Representation now includes L126 alongside F010. The other 17 topic records and 46 individual records retain their exact v14 boundaries. The complete Representation factor is $60,472/82,152 = 0.736099$: written expansion, not achieved whole-topic compression.

For each topic the complete written factor is $\sum_i c(S_i)/(c(U)+\sum_i c(A_i))$. Whole common theory and all complete applications are paid once within that topic. Each row retains its independently adopted nonwhitespace-character coordinate; there is no cross-topic discount, pooled factor or inferred minimum proof complexity. Concrete software listings are excluded on both sides. Proof and mathematical algorithm prose remain counted.

Topic	Originals	Sources	Common	Applications	Factor
A	8	246,570	66,140	110,576	1.395
B	5	124,906	44,905	69,265	1.094
C	5	118,433	30,437	61,248	1.292
D	3	47,894	15,632	29,855	1.053
E	5	140,286	57,473	32,337	1.562
F	2	69,825	15,876	27,398	1.614
G	2	73,199	30,928	85,207	0.630
Representation	2	60,472	36,715	45,437	0.736
Chamber	2	34,177	15,199	21,254	0.938
Matching	1	37,951	16,732	17,168	1.119
ChiralEnumeration	1	27,594	8,831	23,171	0.862
Operations	3	200,630	58,215	100,379	1.265
Gluing	1	27,081	28,416	11,565	0.677
RankContraction	1	15,530	6,937	10,555	0.888
LocallyProjective	2	26,490	25,498	11,966	0.707
MathieuCensus	1	13,736	5,516	35,862	0.332
RankAugmentation	1	33,604	13,904	14,096	1.200
Mixes	3	156,570	34,757	68,693	1.513

The 18-topic CSV and 48-original CSV preserve exact fractions, complete component lengths and historical observations. Factors below one remain visible. The source maps and separate admissions determine the recovered principal contracts; this numerical overview adds no separate proof or question credit.

Individual comparisons: first 24 originals

After acquiring the *whole* common theory and all actual required mathematical, computational and data premises, the conditional factor is $c(S_i)/c(A_i)$. The complete local application or companion bundle remains charged.

Original	Topic	Source	Application	Factor
L032	A	16,558	8,228	2.012
L034	A	21,433	5,801	3.695
F016	A	43,983	15,436	2.849
F013	A	25,562	6,241	4.096
L035	A	51,019	14,514	3.515
N009	A	36,824	28,130	1.309
L128	A	18,424	8,825	2.088
L012	A	32,767	23,401	1.400
N001	B	14,519	7,371	1.970
N003	B	17,780	9,155	1.942
N008	B	18,009	11,132	1.618
L030	B	22,704	12,286	1.848
L040	B	51,894	29,321	1.770
L088	C	13,166	11,311	1.164
N013	C	27,361	15,815	1.730
L108	C	12,849	10,075	1.275
F014	C	20,577	6,962	2.956
L068	C	44,480	17,085	2.603
L066	D	20,916	10,989	1.903
L074	D	20,661	13,367	1.546
L104	D	6,317	5,499	1.149
L079	E	21,212	1,858	11.417
L083	E	20,904	2,922	7.154
L021	E	29,594	14,452	2.048

Representation pays the complete 13-page shared theory and both applications of four and twelve pages:

$$36,715 + 10,985 + 34,452 = 82,152.$$

The sources total $25,348 + 35,124 = 60,472$ characters. The primary remains calibrated selected text, NFC, excluding whitespace. The finite-normal-pair existence proof is now fully in the common paper; no sibling application is free.

After paying whole U and retained premises, F010 has factor 2.307510; L126 has factor 1.019505. These conditional reuse figures do not undo the expansion of the full proof. L126 additionally requires the complete graph certificate and full 19,326-byte JSON; nine validation programs are optional.

Individual comparisons: remaining 24 originals

Original	Topic	Source	Application	Factor
L056	E	15,871	4,144	3.830
L043	E	52,705	8,961	5.882
L014	F	25,574	11,504	2.223
L007	F	44,251	15,894	2.784
L009	G	49,891	23,544	2.119
L092	G	23,308	61,663	0.378
F010	Representation	25,348	10,985	2.308
L126	Representation	35,124	34,452	1.020
L008	Chamber	20,159	7,571	2.663
N016	Chamber	14,018	13,683	1.024
F015	Matching	37,951	17,168	2.211
L028	ChiralEnumeration	27,594	23,171	1.191
N005	Operations	122,002	49,595	2.460
N006	Operations	55,740	38,479	1.449
N007	Operations	22,888	12,305	1.860
L010	Gluing	27,081	11,565	2.342
L037	RankContraction	15,530	10,555	1.471
L084	LocallyProjective	2,071	4,857	0.426
L095	LocallyProjective	24,419	7,109	3.435
L073	MathieuCensus	13,736	35,862	0.383
L020	RankAugmentation	33,604	14,096	2.384
F005	Mixes	57,753	31,077	1.858
L016	Mixes	35,506	15,125	2.348
F007	Mixes	63,311	22,491	2.815

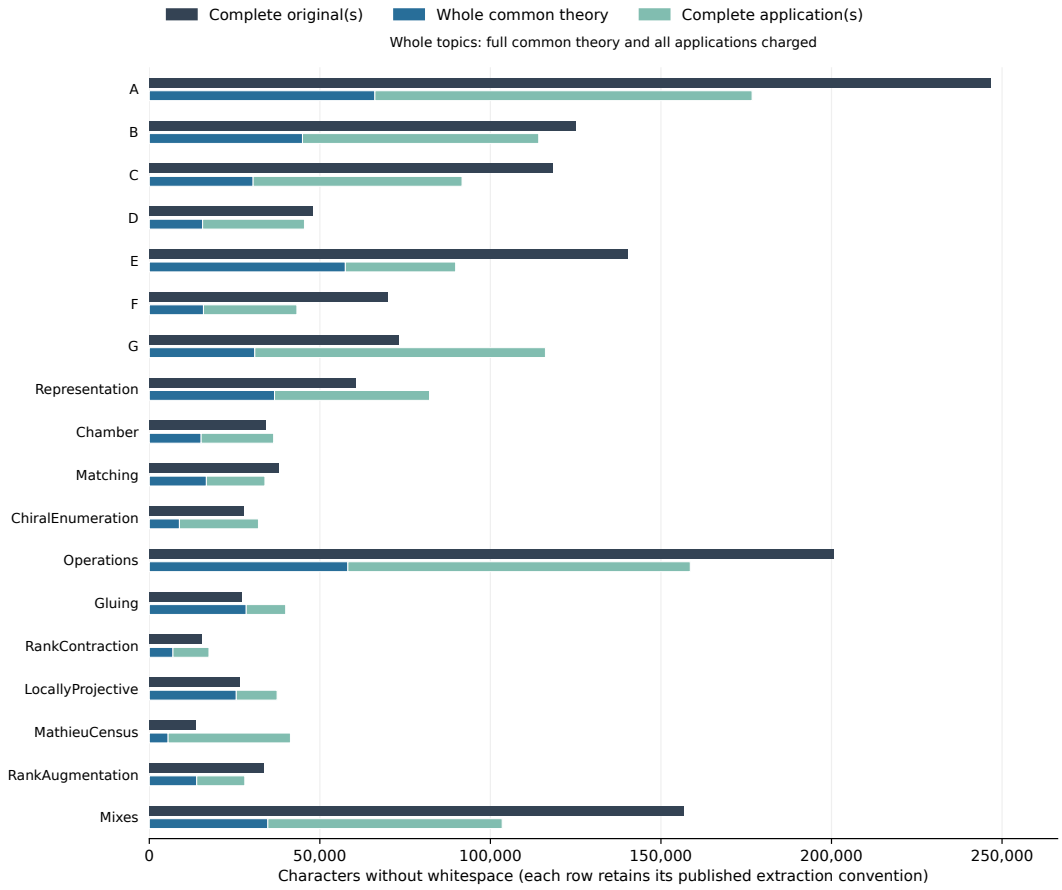
The Representation source contract retains six F010 blocks and 31 L126 dispositions, with finite-field/small-case corrections and an explicit chamber-connectivity hypothesis for splitting. The broader Section 6.2 and Section 8 programmes remain unresolved. The exact source editions and independent proof/input/numerical records are linked by its admission.

The 16-page program appendix prints all ten programs. Its 35,723 characters and the raw programs' 34,823 are alternatives, never added twice. The mandatory JSON contributes 19,315 raw nonspace characters separately; it is not replaced by the appendix. Two current helpers and four partial Lean modules/22 declarations are separate inventories. Neither full source is formalized.

Historical software, retained external proofs, complete graphics and full formalization costs remain unknown. All software-inclusive and pooled factors are null. Operations retains its separate text-only sensitivity; expanded E retains its accepted source qualifications and the source's Magma/L_2-quotient attribution. Prior overviews remain historical records.

Whole topics

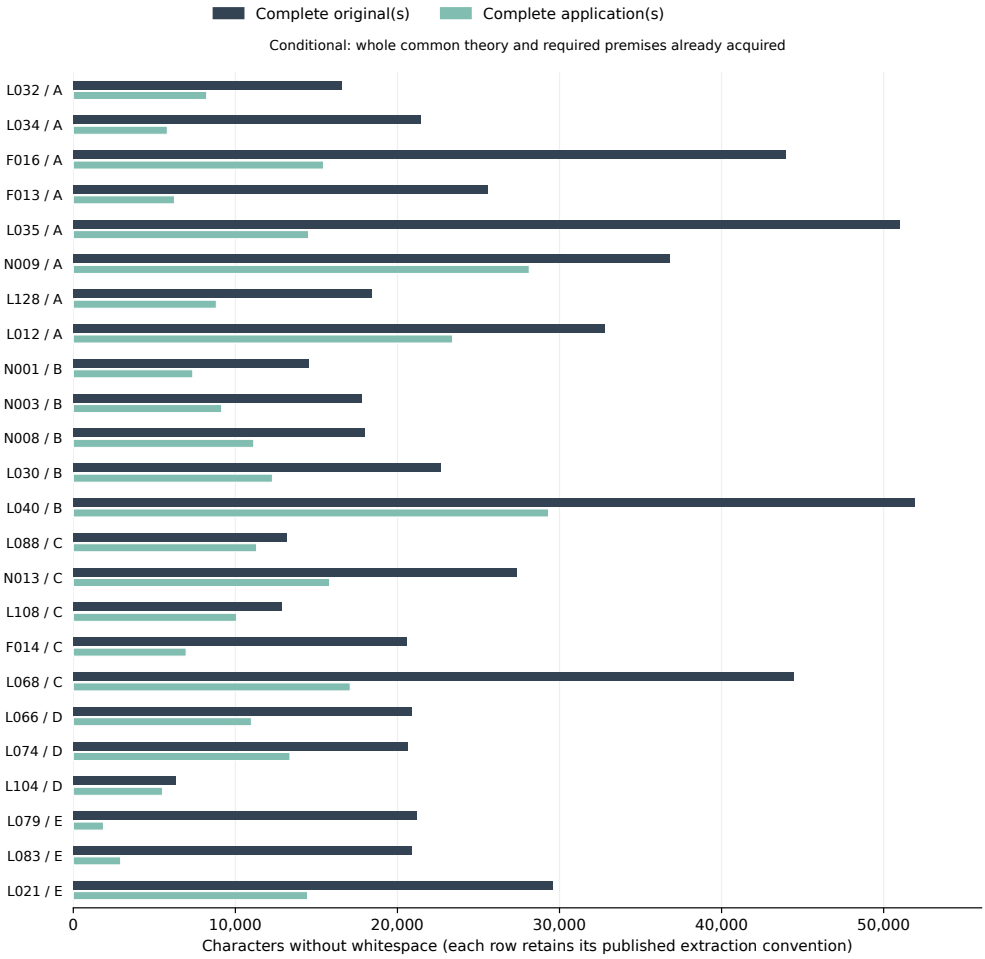
Grey bars show the complete original manuscript text in each row's adopted coordinate. Replacement bars stack the entire common theory and all applications. The axis starts at zero; the tables give the ratios.



The 54 bars account for all 18 topics. No necessary finite certificate or external premise becomes optional merely because software and retained external proof costs are outside the manuscript coordinate.

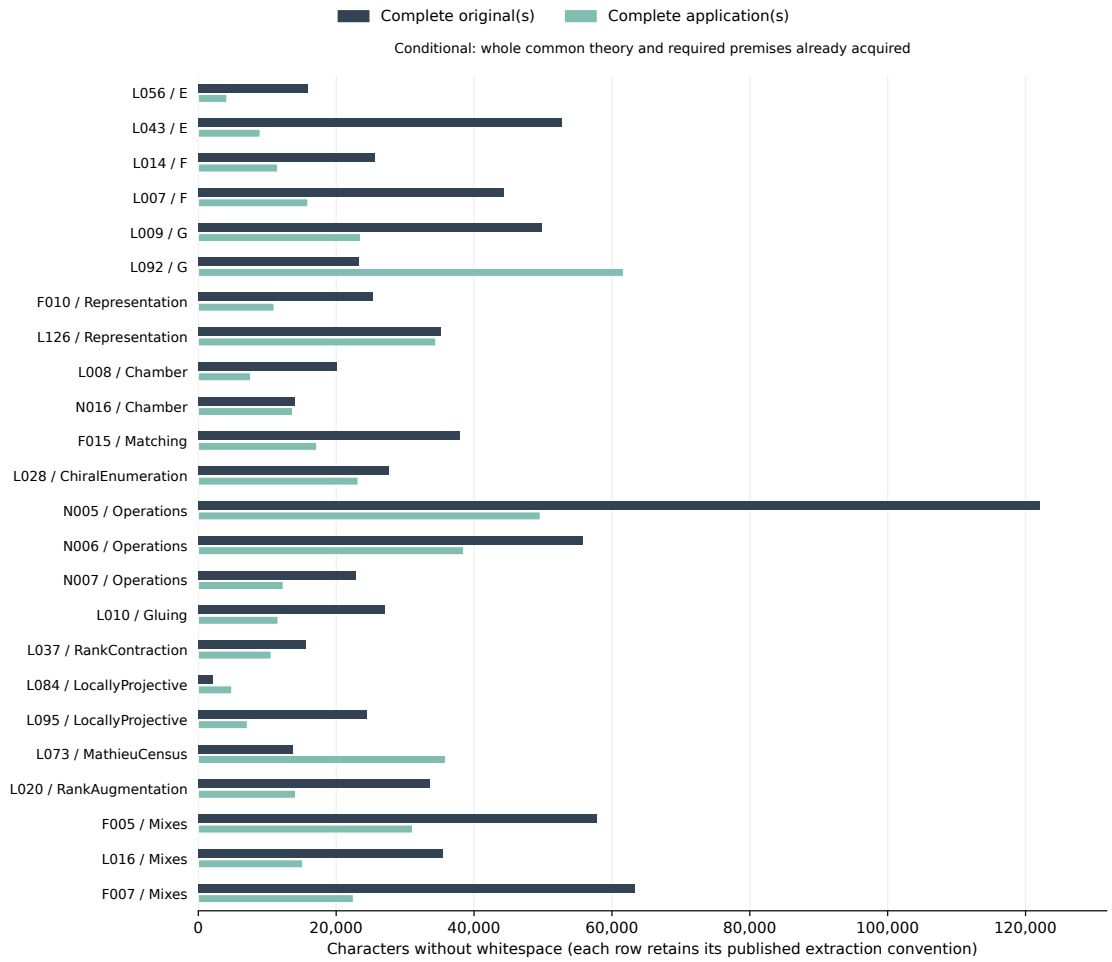
Conditional lengths: first 24 originals

Whole common theory and required premises are acquired in advance. Grey and teal bars compare complete source and local application text; the whole-topic table still pays the common theory.



Conditional lengths: remaining 24 originals

All complete companion papers remain charged. The source order is documentary, not an optimal acquisition order or a claimed sawtooth.



This edition supplies 66 exact written comparisons and 150 plotted bars. The active programme remains 106 originals, with 48 recovered and 58 pending. Only L126 is newly reflected from its separate scientific admission; F010 was already admitted. No new question or priority claim is made.